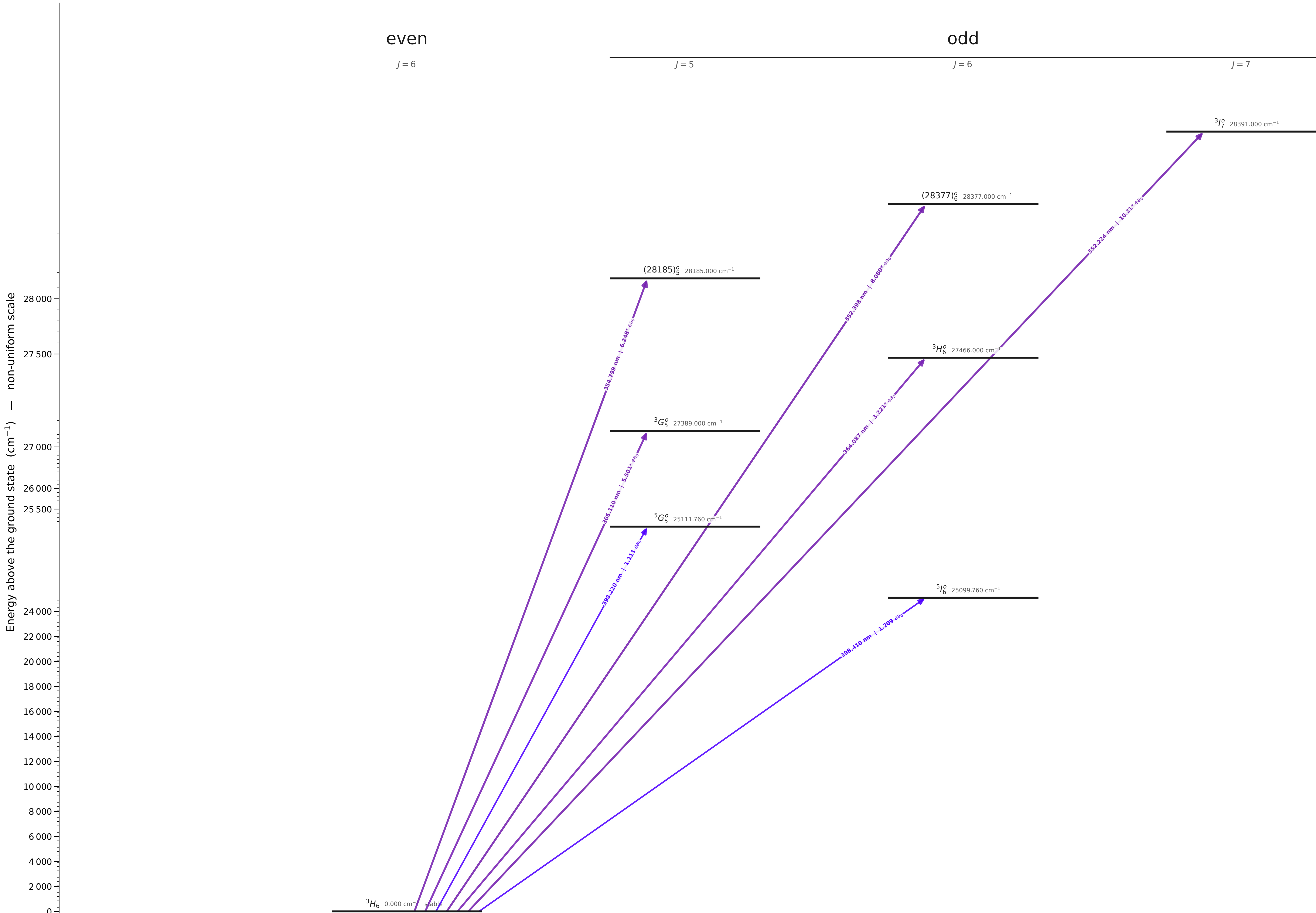


# Fermium

Fm (Fm I). Classified transitions below 2 μm (levels and lines from the literature): 8 levels, 7 lines.

Fm<sup>+</sup> ionisation limit 52400.00 cm<sup>-1</sup> = 6.49677 eV (190.840 nm)



## Key transitions

| Transition                                                             | λ vac (nm) | λ air (nm) | ν (THz)  | U  er  J')  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|------------------------------------------------------------------------|------------|------------|----------|--------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| <sup>3</sup> H <sub>6</sub> - <sup>3</sup> I <sub>7</sub> <sup>o</sup> | 352.2243   | 352.1236   | 851.1408 | 10.21*                         | 3.223e+08            | -              | -          | ground-state resonance found in the 2005 level search        |
| <sup>3</sup> H <sub>6</sub> - (28377) <sup>o</sup> <sub>6</sub>        | 352.3981   | 352.2973   | 850.7211 | 8.080*                         | 2.325e+08            | -              | -          | ground-state resonance found in the 2005 level search        |
| <sup>3</sup> H <sub>6</sub> - (28185) <sup>o</sup> <sub>5</sub>        | 354.7987   | 354.6973   | 844.9650 | 6.248*                         | 1.610e+08            | -              | -          | ground-state resonance found in the 2005 level search        |
| <sup>3</sup> H <sub>6</sub> - <sup>3</sup> H <sub>6</sub> <sup>o</sup> | 364.0865   | 363.9828   | 823.4100 | 3.221*                         | 3.351e+07            | -              | -          | ground-state resonance found in the 2005 level search        |
| <sup>3</sup> H <sub>6</sub> - <sup>3</sup> G <sub>5</sub> <sup>o</sup> | 365.1101   | 365.0061   | 821.1016 | 5.501*                         | 1.145e+08            | -              | -          | ground-state resonance found in the 2005 level search        |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.2198   | 398.1072   | 752.8316 | 1.111                          | 3.600e+06            | -              | -          | resonance R2: first step of two-step resonance ionization... |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> I <sub>6</sub> <sup>o</sup> | 398.4102   | 398.2975   | 752.4719 | 1.209                          | 3.600e+06            | -              | -          | resonance R1: first step of two-step resonance ionization... |

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>255</sup>Fm hyperfine structure (I = 7/2)

| Level                                    | A (MHz)    | B (MHz)    | Shift of each F level from the centre of gravity (MHz) |                 |                 |                  |                  |                  |                  |                  |  |  |
|------------------------------------------|------------|------------|--------------------------------------------------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|--|--|
| <sup>3</sup> H <sub>6</sub>              | -149.5(11) | -10460(40) | F=5/2: -497.48                                         | F=7/2: +762.23  | F=9/2: +1974.32 | F=11/2: +2833.14 | F=13/2: +2965.12 | F=15/2: +1928.77 | F=17/2: -785.32  | F=19/2: -5754.50 |  |  |
| <sup>5</sup> I <sub>6</sub> <sup>o</sup> | -309.2(12) | -12790(40) | F=5/2: +2488.47                                        | F=7/2: +3586.38 | F=9/2: +4499.67 | F=11/2: +4854.61 | F=13/2: +4194.40 | F=15/2: +1979.22 | F=17/2: -2413.84 | F=19/2: -9690.70 |  |  |
| <sup>5</sup> G <sub>5</sub> <sup>o</sup> | -9.7(13)   | -13570(50) | F=3/2: -5676.63                                        | F=5/2: -3600.76 | F=7/2: -1146.88 | F=9/2: +1232.68  | F=11/2: +2956.36 | F=13/2: +3313.33 | F=15/2: +1463.56 | F=17/2: -3562.25 |  |  |

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |U||er||J')| in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

## Isotope shifts

| Transition                                                             | λ vac (nm) | Isotopes | Shift (MHz)  |
|------------------------------------------------------------------------|------------|----------|--------------|
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 245-250  | 30400(2200)  |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 246-250  | 21000(1700)  |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 248-250  | 9500(1000)   |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 249-250  | 5800(3400)   |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 254-250  | -27000(1400) |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 255-250  | -31500(900)  |
| <sup>3</sup> H <sub>6</sub> - <sup>5</sup> G <sub>5</sub> <sup>o</sup> | 398.220    | 257-250  | -39500(2400) |

Shift = ν(first isotope) - ν(second isotope).